PHY-112 | Principles of Physics-2

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Spring 2025 | Class #10

Department of Mathematics & Natural Sciences

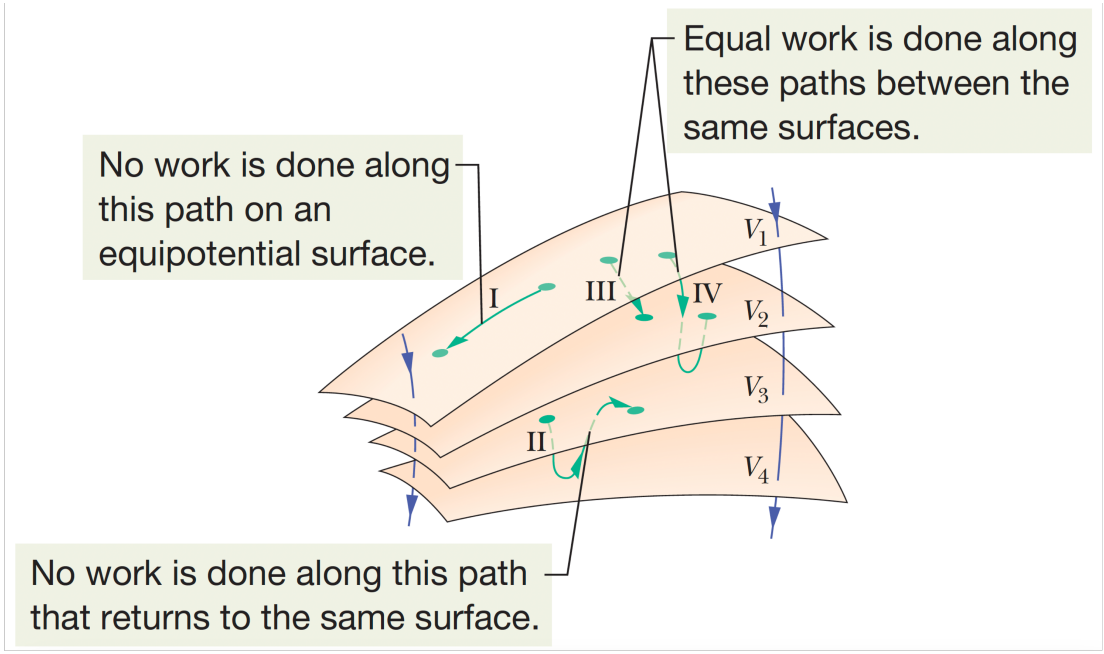
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The Beauty of Equipotential Surfaces

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What are Equipotential Surfaces?
Why should you care?

Adjacent points around an electric field source that have the same electric potential form an equipotential surface, which can be either an imaginary surface or a real, physical surface.

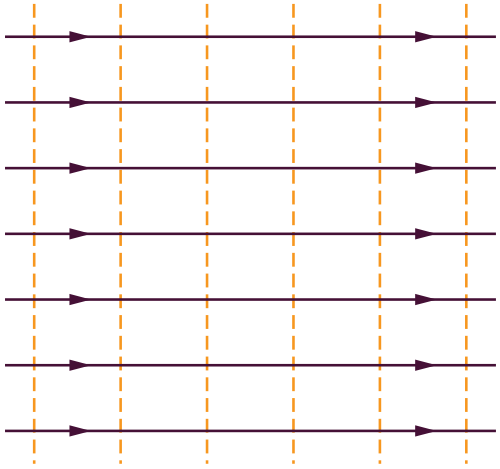


No net work W is done on a charged particle by an electric field when the particle moves between two points i and f on the same equipotential surface.

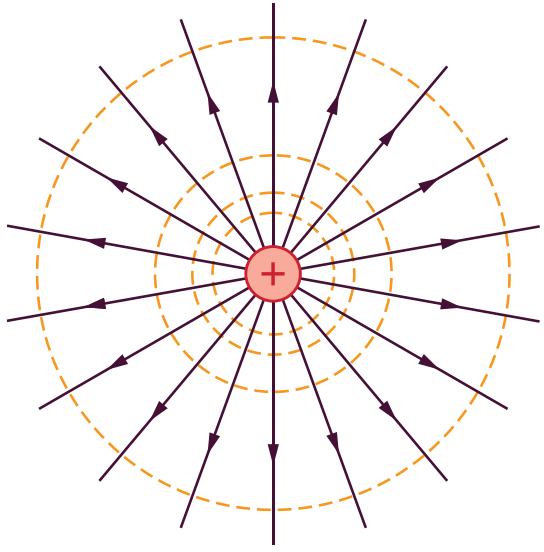
How do Equipotential Surfaces look like?

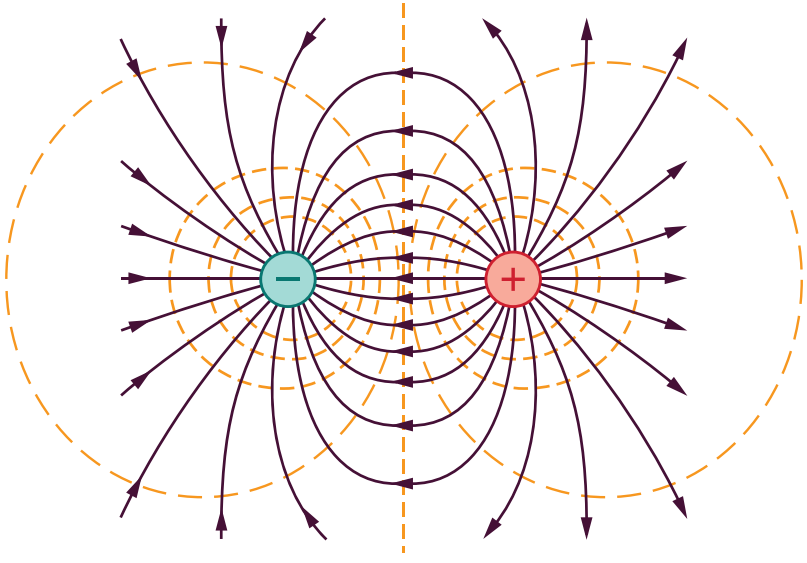
Some familiar sources

(a) Uniform field source

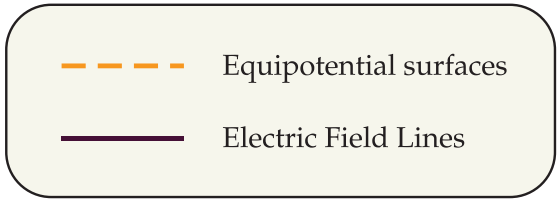


(b) Positive point source (Non-Uniform)





(c) Dipole source (Non-Uniform)



The Geometry of Electric Fields

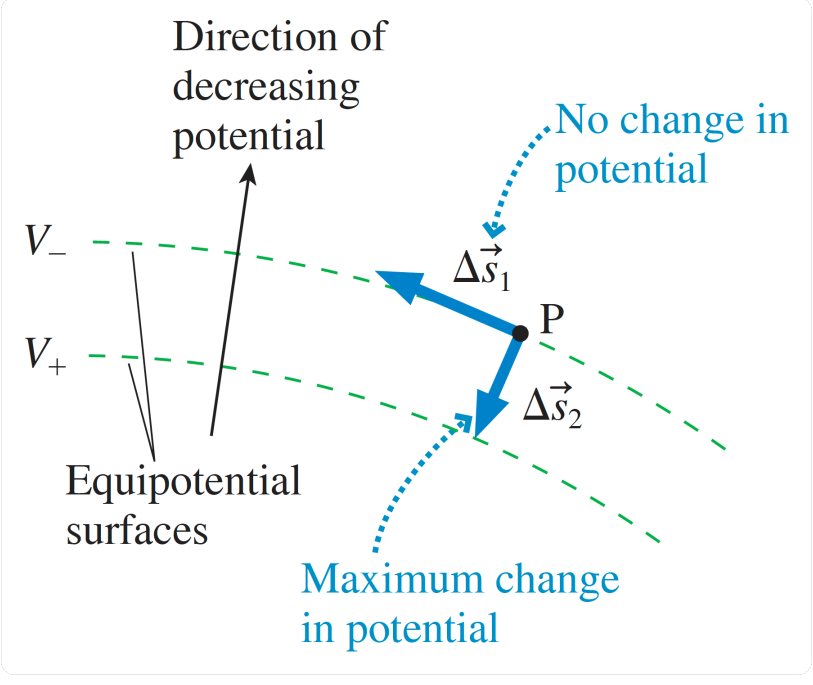
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Geometry of Electric Potential and Field

How to diagnose $\vec{E}$ from Potential

$$\vec{E} = -\vec{\nabla}V = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right) = -\left(E_x\hat{i} + E_y\hat{j} + E_z\hat{k}\right)$$

$$E_x = -\frac{\partial V}{\partial x}, E_y = -\frac{\partial V}{\partial y}, E_z = -\frac{\partial V}{\partial z}$$



Incepting Ideas (1)

Hint: Follow the decreasing potential path in each direction

Q: The electric potential at points in an xy plane is given by

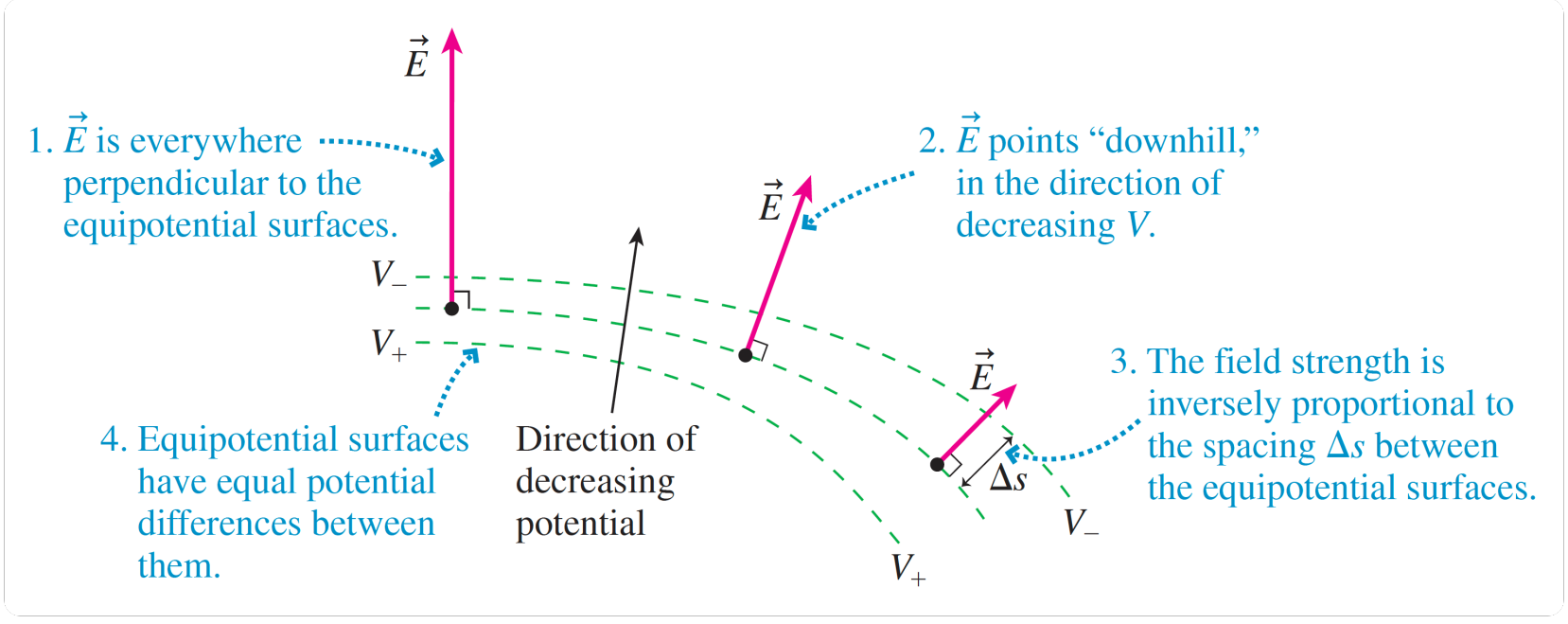
$$V = \left(2.0\,\mathrm{V\,m^{-2}}\right) x^2 - \left(3.0\,\mathrm{V\,m^{-2}}\right) y^2$$

What are the magnitude and direction of the electric field at the points (3.0 m, 2.0 m), (0.0 m, −2.0 m) and (2.0 m, 2.0 m)?

Geometry of Electric Potential and Field

How to diagnose $\vec{E}$ from a diagram

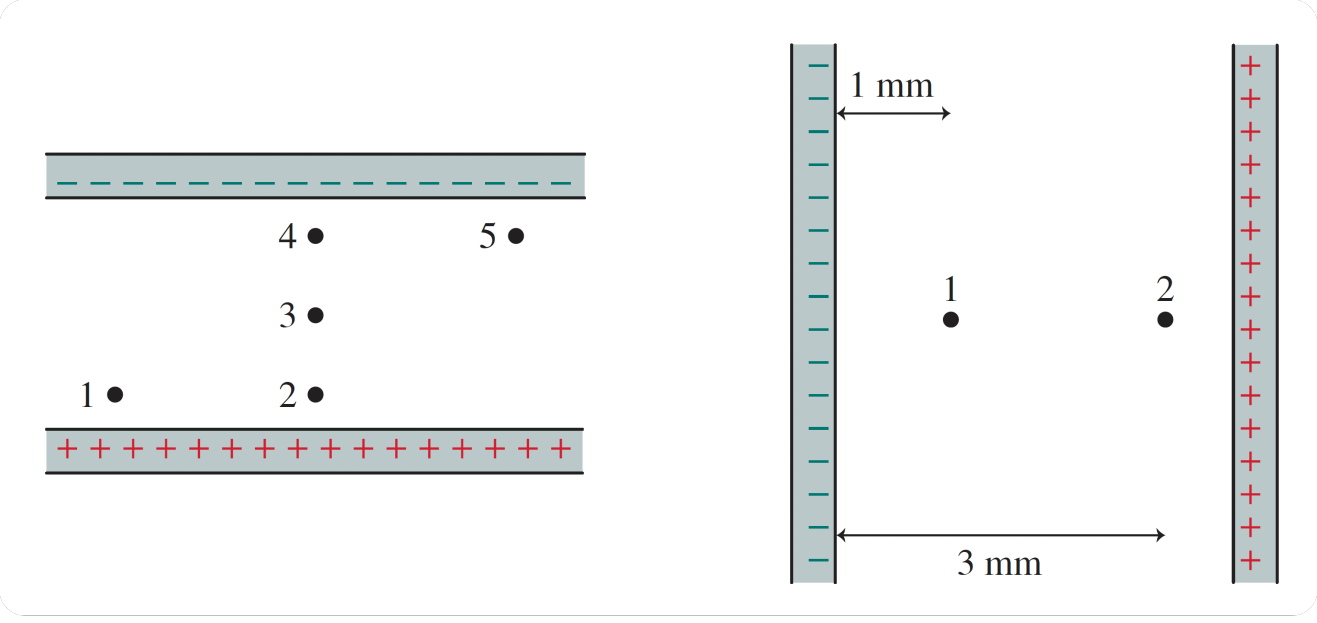
$\vec{E}$-field changes (perpendicular) between two equipotential surfaces toward decreasing potential.



Incepting Ideas (2)

Hint: Follow the potential change

Q: (Left) Rank in order, from largest to smallest, the electric potentials V_1 to V_5 at points 1 to 5.

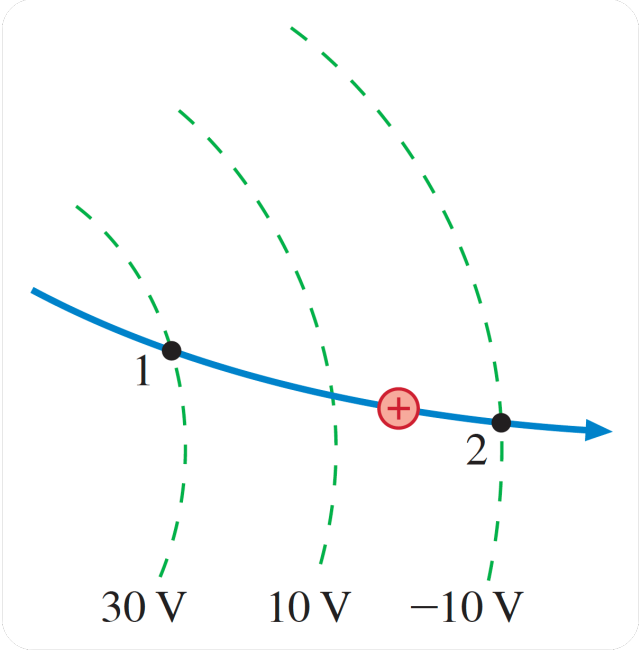


Q: (Right) In the right diagram, two points inside a capacitor are spotted. Let $V = 0$ V at the negative plate. Find $V_1 : V_2$ and $E_1 : E_2$.

Incepting Ideas (3)

Hint: The fall in ΔU is the exact opposite of gain in ΔK

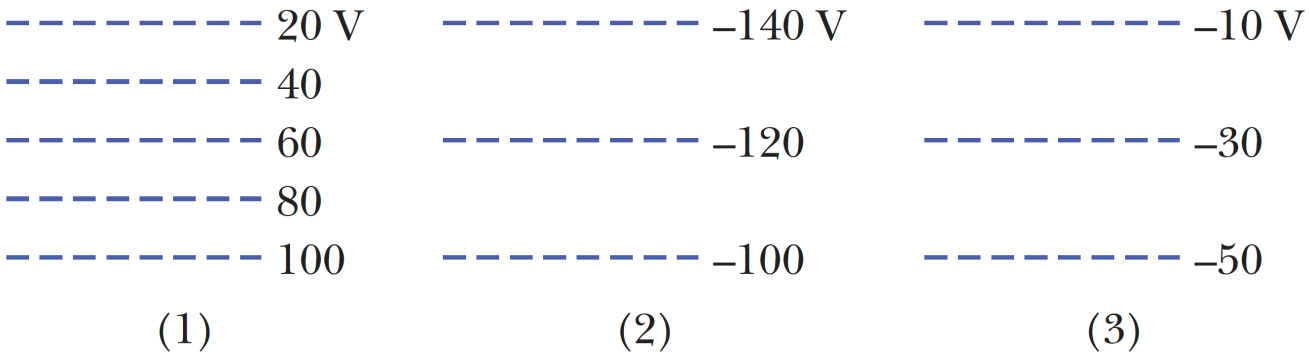
Q: A proton's speed as it passes point 1 is 50 km s^{-1}. It follows the trajectory shown in the diagram below. What is the proton's speed at point 2?



Incepting Ideas (4)

Hint: Follow the decreasing potential path

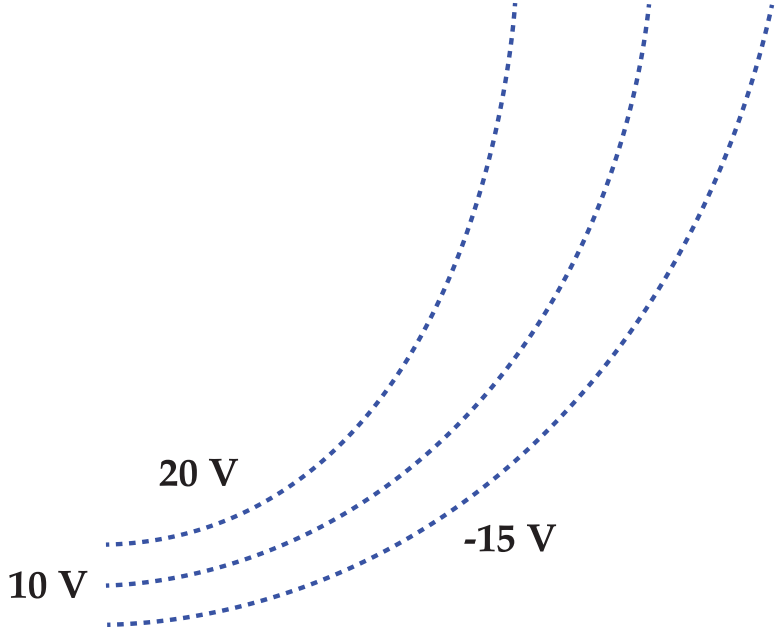
Q: Three sets of cross sections of equipotential surfaces in uniform electric fields; all three cover the same size region of space. The electric potential is indicated for each equipotential surface. (a) Rank the arrangements according to the magnitude of the electric field present in the region, greatest first. (b) In which is the electric field directed down the page?



Incepting Ideas (5)

Hint: Follow the decreasing potential path

Q: Three equipotential surfaces (dashed lines) are shown. Draw an accurate diagram of the electric field intensity and direction for this setup. The diagram must be scaled and properly labeled. Also, give only one line of explanation on why You chose such (whatever You chose) specific direction for $\vec{E}$.



The Electric Force Is a Conservative Force

It conserves the mechanical energy of the system

Potential energy can only be defined only if the force is CONSERVATIVE.

$$F_E = -\frac{\partial U_{\text{elec}}}{\partial r}$$

This means that the work done on the charged particle as it moves from position i to position f is **independent of the path** followed between i and f.

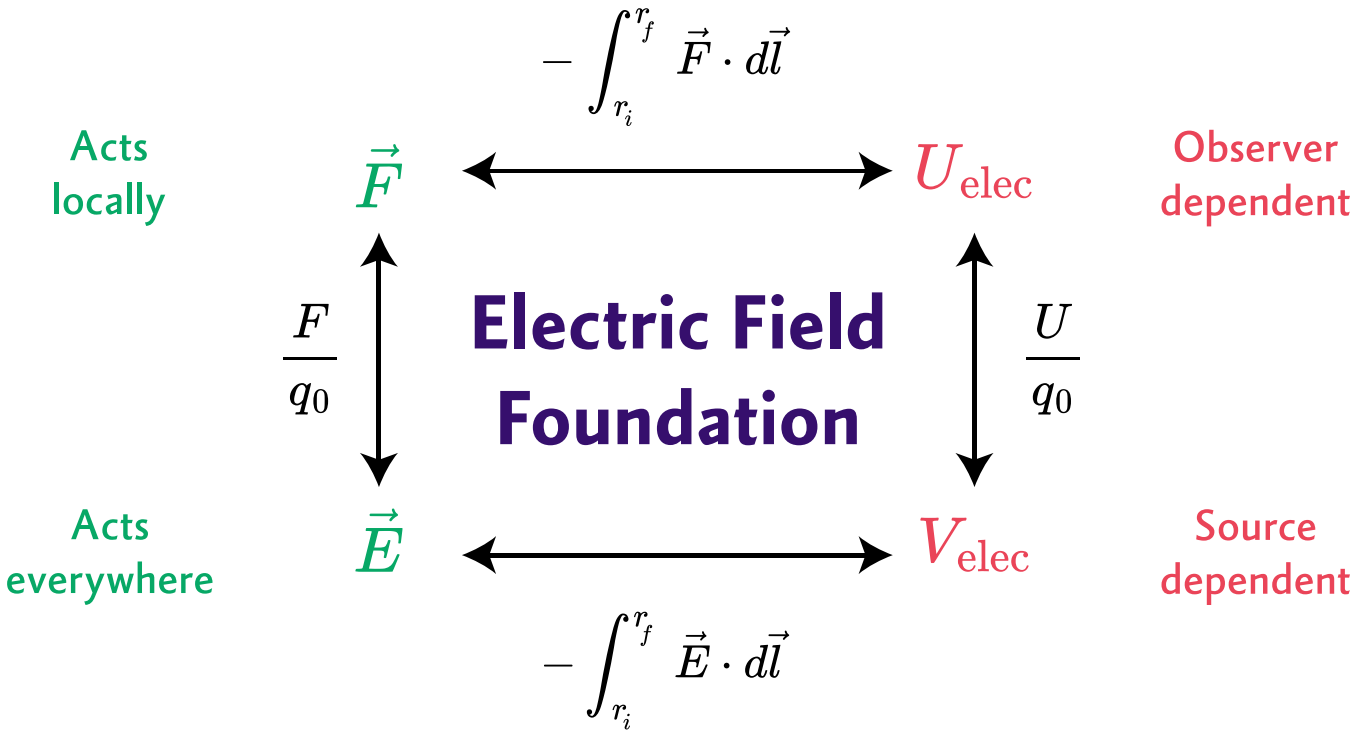
$$W_{i\to f} = -W_{f\to i}$$

What does the force conserve? $\longrightarrow$ **Mechanical Energy** of the system

Where are we now?

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Looking at the Big Picture: Where We Are Now
Der Anfang ist das Ende und das Ende ist der Anfang



That is it for today!

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